WIRESHARK SOC LAB

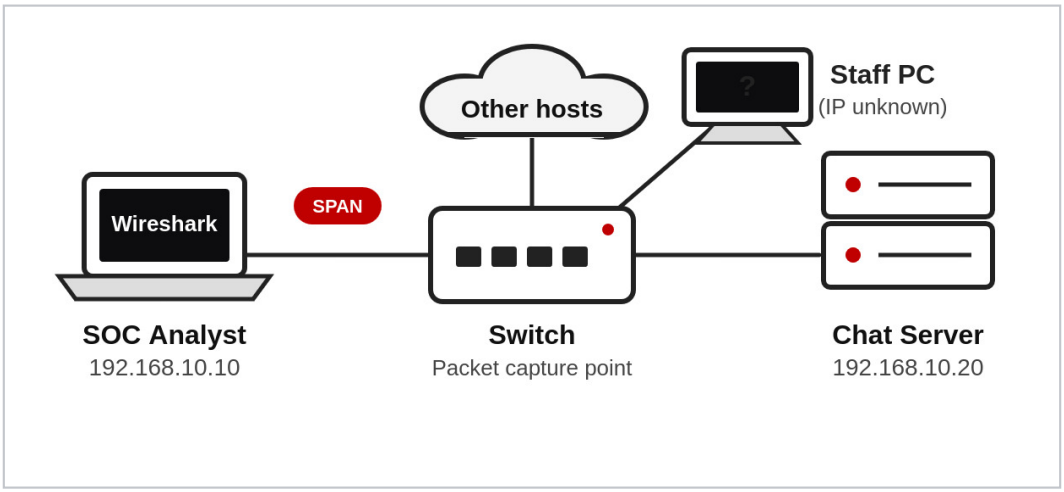
Follow TCP Stream

Online lab: https://networkwalks.com/wireshark-lab4/

SCENARIO

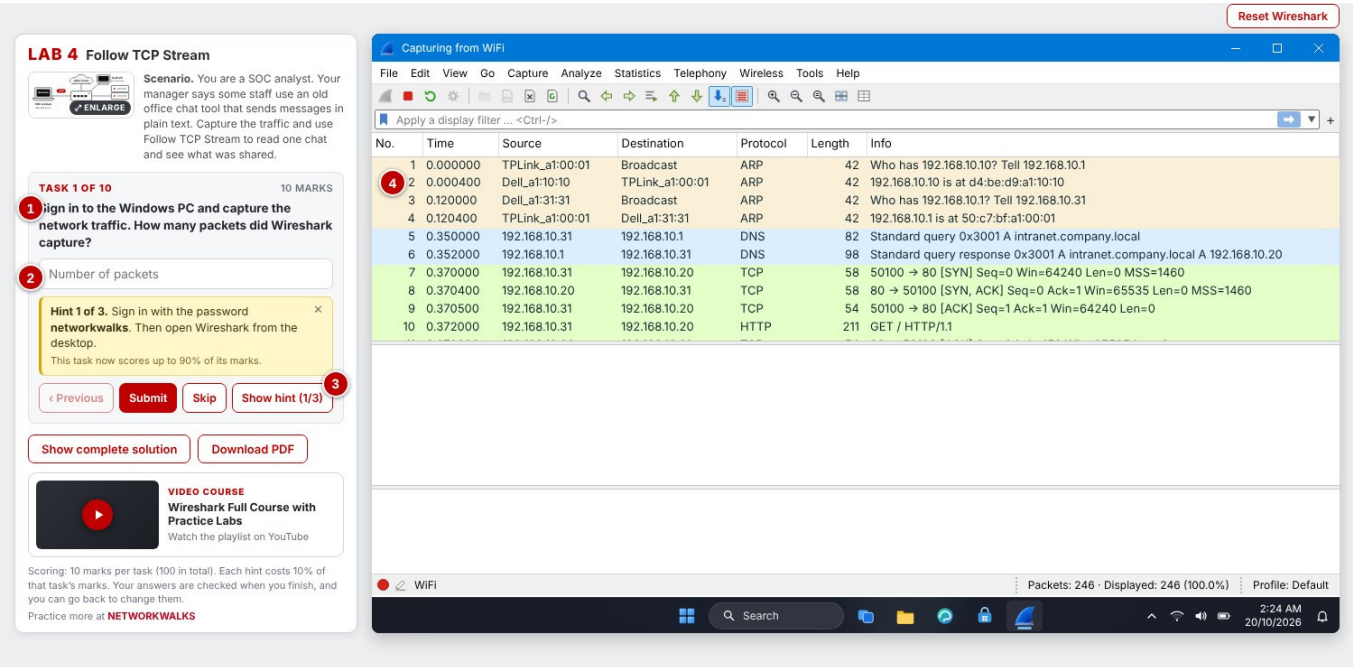
You are a SOC analyst. Your manager says some staff use an old office chat tool that sends messages in plain text. Capture the traffic and use Follow TCP Stream to read one chat and see what was shared.

TOPOLOGY DIAGRAM



LAB INSTRUCTIONS

How to complete this lab



The lab screen: the task box on the left, the Windows 11 PC with Wireshark on the right.

1. Read the scenario and the task in the left box (1). Do one task at a time.
2. On the right PC, sign in with the password hint. Double-click Wireshark, then double-click the WiFi interface to start the capture (4).
3. Use the filter bar, the packet list and the Statistics menu to find the answer for the task.
4. Type your answer in the box (2), then press Submit. Use Skip to leave a task for later and Previous to go back. Answers are checked at the end.
5. Stuck? Press Show hint (3). Each hint costs 10% of the marks of that task, only the first time you open it. A task with no answer scores 0.
6. On task 10 press Submit, add a comment if you like, then press Finish. You get a result card that you can download as a picture and share. Show complete solution is always available.

TASKS

Sign in to the Windows PC (user Student, the password hint is on the sign-in screen), start a capture in Wireshark and complete the 10 tasks below. Each task is worth 10 marks (100 in total). Each hint costs 10% of that task's marks.

Task 1. Capture the traffic

10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Hint 1: Sign in with the password networkwalks. Then open Wireshark from the desktop.
Hint 2: Double-click the interface that shows an activity graph.
Hint 3: Read the number after Packets: in the status bar at the bottom of the Wireshark window.

Answer: 246

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 246.

Task 2. Find the chat port

10 marks

Most of this traffic is web traffic on port 80 or 443. One computer also used a plain text office chat tool on a different TCP port. Which port is it?

Hint 1: Open Statistics › Conversations and click the TCP tab.
Hint 2: Look at the Port B column. Most rows show 80 or 443. One row shows a different number.

Answer: 5000

Solution: Open Statistics › Conversations and click the TCP tab. Most rows use port 80 or 443. One conversation uses port 5000. This is the office chat.

Task 3. Who started the chat?

10 marks

The chat uses TCP port 5000. Which IP address started the chat? (The computer that sends the first SYN packet starts the connection.)

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter.
Hint 2: Look at the first packet in the list. It is a SYN packet. Read its Source column.

Answer: 192.168.10.45

Solution: Type

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tcp.port == 5000
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in the filter bar. The first packet is a [SYN]. Its Source is 192.168.10.45. This computer started the chat with the server 192.168.10.20.

Task 4. Stream number

10 marks

Follow the chat with Follow › TCP Stream. The title of the window shows tcp.stream eq and a number. What is the stream number of the chat?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: Read the number after tcp.stream eq in the title of the Follow window.

Answer: 5

Solution: Type

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tcp.port == 5000
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in the filter bar, right-click a packet that has data and choose Follow › TCP Stream. The window title shows tcp.stream eq 5.

Task 5. Size of the chat

10 marks

Look at the bottom bar of the Follow TCP Stream window. How many bytes does the entire conversation have?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: Read the number in Entire conversation (… bytes) at the bottom of the window.

Answer: 413

Solution: Open Follow › TCP Stream on the chat. The bottom bar shows Entire conversation (413 bytes).

Task 6. Who needs help?

10 marks

Read the chat. What is the first name of the staff member who asks for help?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: Read the first message in red text. The person tells who they are.

Answer: Ali

Solution: In the Follow window, the first red message says “Hi, this is Ali from Accounts.” The name is Ali.

Task 7. The problem drive

10 marks

Which shared drive could the staff member not open?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: Read the red messages. The person tells which drive has the problem.

Answer: Finance

Solution: In the chat, Ali says “The Finance drive.” The drive is Finance.

Task 8. Temporary password

10 marks

IT gave the staff member a temporary password in the chat. What is it?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: Read the blue messages. IT writes the password after the words “temporary password is”.

Answer: Tiger#4821

Solution: In the blue text, IT writes “Your temporary password is Tiger#4821”. The password is Tiger#4821. This is why chat tools must be encrypted.

Task 9. Guest Wi-Fi password

10 marks

The staff member also asked for the guest Wi-Fi password. What is it?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: Read the blue messages. IT writes the guest Wi-Fi password after the words “Guest Wi-Fi password:”.

Answer: BlueHarbor77

Solution: In the blue text, IT writes “Guest Wi-Fi password: BlueHarbor77”. The password is BlueHarbor77.

Task 10. Text colour

10 marks

In the Follow TCP Stream window, what colour is the text sent by the computer that started the chat (the client)?

Hint 1: Type tcp.port == 5000 in the filter bar and press Enter. Right-click a packet that has data (Len above 0) and choose Follow, then TCP Stream.
Hint 2: The client is Ali, who sends the first message after the welcome line. Look at the colour of Ali’s messages.

Answer: Red

Solution: The client (192.168.10.45, Ali) sends the red text. The server sends the blue text.

SUMMARY

Chat:	192.168.10.45 to server 192.168.10.20
TCP port:	5000 (stream 5)
Shared:	Temporary password and guest Wi-Fi password
Lesson:	Use encrypted tools (TLS/HTTPS)

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro